AI-Powered Smart Service Marketplace for Household Maintenance in Sri Lanka

R26-IT-090

Project Proposal Report

SHIWANTHIKA M.M – IT22283658

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
~~IT22283~~ Shiwanthika M.M	IT22283658	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof.Dasuni Nawinna	20/3/2026	

ABSTRACT (Maximum 300 words)

Finding skilled and trustworthy household service workers is challenging in developing countries due to informal labor markets, limited records, and reliance on word-of-mouth. Existing job recommendation research focuses on corporate platforms, leaving intelligent matching systems for household services underexplored.

This research addresses the gap by proposing an Intelligent Worker Recommendation Engine designed specifically for physical home maintenance services. The system automatically identifies and ranks the most suitable worker for a given task by analyzing multiple factors, including skill compatibility, geographical proximity,education qualifications,worker category,past performance ratings,job compeation, and availability. A weighted multi-factor scoring model is applied to generate a Best-Fit Score for each worker, enabling accurate and fair recommendations.

The expected outcome is a transparent and efficient worker recommendation system that boosts reliability and platform performance, promotes fair job distribution and community trust, and improves customer satisfaction, operational efficiency, and scalability.

Keywords:Intelligent Worker Recommendation,Household Services,Skill Matching,Best-Fit Score,Worker Performance

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to my research supervisor, [Supervisor's Name], for their invaluable guidance, constant encouragement, and technical insights throughout the development of this research proposal. Their expertise in machine learning and market dynamics was instrumental in shaping the "AI-Powered Smart Service Marketplace for Household Maintenance in Sri Lanka"

I am also grateful to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary academic environment and resources to pursue this study. My thanks also extend to the various household maintenance service providers and technicians in Sri Lanka who participated in my preliminary market surveys; their honest feedback provided the essential baseline data for my labor and transport rate analysis.

Furthermore, I would like to thank my colleagues and friends for their constructive criticism and support during the brainstorming phases of this project. Finally, a special word of thanks to my family for their continuous motivation and belief in my work.

[Shiwanthika M.M]

[20/03/2026]

TABLE OF CONTENT (Auto Generated)

Table of Contents

DECLARATION 2

ABSTRACT (Maximum 300 words) 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT (Auto Generated) 5

LIST OF TABLES 6

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 8

1. INTRODUCTION 9

1.1 Background and Context 9

1.2 Literature Review 10

1.3 Research Gap 12

2. OBJECTIVE 14

2.1 Main Objective 14

2.2 Specific Objectives 14

3. METHODOLOGY 15

4. HIGH-LEVEL SYSTEM ARCHITECTURE 18

5. USER REQUIREMENTS 20

6. COMMERCIALIZATION PLAN 23

7. BUDGET AND JUSTIFICATION 26

8. WORK BREAKDOWN STRUCTURE (WBS) 27

9. GANTT CHART 31

10. REFERENCES LIST 32

11. APPENDICES 34

LIST OF TABLES

Table 1: List of Abbreviation……...……8

Table 2: Budget Allocation chart……..……...…23

Table 3: Timeline Alignment chart……25-26

LIST OF FIGURES

Figure 1: Component specific system diagram 16

Figure 2: System Architecture Diagram..... 17

Figure 3: Gantt Chart Diagram.....27

LIST OF ABBREVIATIONS

Abbreviation	Description
IWR	Intelligent Worker Recommendation
ML	Machine Learning
RF	Random Forest
API	Application Programming Interface
SRS	Software Requirement Specification
SDG	Sustainable Development Goal
DB	Database
UI	User Interface

1. INTRODUCTION

1.1 Background and Context

Household maintenance services such as plumbing, electrical repair, carpentry, and appliance maintenance are essential for everyday living. However, finding reliable and skilled workers for these services remains a major challenge in many countries, including Sri Lanka. Homeowners often rely on personal contacts, social media posts, or informal recommendations to find workers. This process is time-consuming, unreliable, and sometimes unsafe because users cannot easily verify a worker's skills, experience, or reputation. As urban populations grow and lifestyles become busier, the demand for fast and trustworthy service connections is increasing. Therefore, developing intelligent digital solutions to match service requests with suitable workers has become increasingly important.

The lack of an efficient worker-matching system affects multiple groups of people. Homeowners face difficulties in quickly finding trustworthy service providers, especially during urgent situations such as plumbing leaks or electrical failures. At the same time, many skilled workers struggle to find consistent job opportunities because they lack proper platforms to showcase their skills and reach potential customers. According to global labour reports from the International Labour Organization, millions of skilled informal workers operate outside structured digital platforms, which limits their visibility and access to job opportunities. Locally, the growth of urban housing and service demand in Sri Lanka has increased the need for more organized service marketplaces that connect customers with qualified workers efficiently.

Several existing solutions attempt to address this issue through online service marketplaces and gig-economy platforms. Examples include digital platforms that allow users to search for nearby workers or post service requests. However, many of these systems rely on simple filtering methods such as location or rating. They often fail to consider multiple important factors simultaneously, including worker skills, past performance, availability, and proximity. As a result, users may receive irrelevant or low-quality recommendations, leading to inefficient service matching. Additionally, many platforms lack intelligent algorithms capable of analyzing user needs and worker profiles in a dynamic and data-driven way.

To overcome these limitations, this project proposes the development of an Intelligent Worker Recommendation Engine as part of a smart service marketplace. The proposed solution aims to analyze user service requests and recommend the most suitable workers using a data-driven approach. The system will consider multiple attributes such as worker skills, experience, ratings, availability, and geographic distance. By applying recommendation algorithms and scoring mechanisms, the engine will rank workers according to their suitability for a given task and present the best options to users. This approach is expected to improve service matching efficiency, reduce search time for users, and create more job opportunities for skilled workers.

The proposed system also aligns with global sustainable development initiatives, particularly United Nations Sustainable Development Goals (SDGs). Specifically, it supports Sustainable Development Goal 8: Decent Work and Economic Growth by creating digital opportunities for

skilled workers to access more jobs and fair income. Additionally, it contributes to Sustainable Development Goal 9: Industry, Innovation and Infrastructure by promoting technology-driven solutions that improve service marketplaces and digital infrastructure.

From a research perspective, this work aligns with the Artificial Intelligence and Data Science research cluster, which focuses on intelligent decision-making systems, recommendation algorithms, and data-driven platforms. By integrating machine learning concepts and recommendation techniques, the Intelligent Worker Recommendation Engine aims to demonstrate how AI-based systems can improve service marketplaces and enhance the efficiency of worker-customer matching processes in modern digital economies.

1.2 Literature Review

Recent studies have shown a growing interest in intelligent recommendation systems that match workers with suitable tasks in digital marketplaces and crowdsourcing environments. These systems aim to improve efficiency by automatically identifying workers who possess the appropriate skills, experience, and reliability to complete specific tasks. Several research approaches have been proposed, including collaborative filtering, context-aware recommendation models, machine learning algorithms, and graph-based methods.

One widely studied approach in worker recommendation systems is collaborative filtering, which analyzes the interactions between users and tasks to predict suitable matches. For example, research on improved collaborative filtering algorithms demonstrates that worker-task similarity matrices can be used to recommend tasks more efficiently by analyzing historical interaction data between workers and previously completed tasks. This method improves the efficiency of task searching and resource utilization in crowdsourcing platforms. However, traditional collaborative filtering methods often suffer from problems such as data sparsity and the cold-start problem when new workers or tasks have little historical data.[4]

Another research direction focuses on learning worker preferences and reliability through machine learning techniques. A study on task recommendation using a Multi-Armed Bandit (MAB) framework proposes a model that learns worker preferences and reliability over time to recommend tasks that workers are more likely to complete successfully. This approach allows the system to continuously improve recommendations based on feedback and performance data. Despite its effectiveness, the approach requires continuous training and sufficient interaction data, which may not always be available in early-stage platforms.[1]

Several studies have also explored context-aware recommendation systems for crowdsourcing environments. For instance, temporal context-aware task recommendation models utilize probabilistic matrix factorization to analyze worker-task relationships and identify latent features that influence task suitability. These systems consider contextual factors such as time, task difficulty, and worker activity patterns to improve recommendation accuracy. However,

the complexity of these models can increase computational costs and make real-time implementation challenging in large-scale systems.[8]

Recent research has further investigated multi-dimensional worker modeling techniques. The GRBMC (Group Recommendation Based on Multi-Community collaboration) approach analyzes worker characteristics such as ability, activity level, preferences, and community relationships to recommend suitable worker groups for tasks. By considering multiple attributes rather than a single metric, the model improves the quality of task allocation and reduces inefficiencies in crowdsourcing platforms. Nevertheless, such approaches often require large datasets and complex graph analysis, which can be difficult to implement in small-scale service platforms.[6]

Other researchers have explored clustering and role-based recommendation techniques. In role-based clustering models, workers are grouped according to their roles, preferences, and performance patterns, allowing the system to recommend tasks more effectively within each cluster. This method reduces the complexity of large worker-task datasets and improves recommendation accuracy. However, clustering approaches may struggle to adapt quickly when worker preferences or skills change over time.[9]

More advanced research has introduced deep learning and knowledge graph approaches for worker recommendation. Hybrid neural network models combined with knowledge graphs can capture complex relationships between workers and tasks by analyzing semantic connections, preferences, and historical performance. Experimental results show that these models can significantly improve recommendation accuracy compared to traditional methods. However, they require extensive training data and high computational resources, which may not be practical for smaller systems.[10]

In addition to academic research, several commercial platforms have implemented service recommendation mechanisms in real-world applications. Online gig marketplaces such as Upwork, Fiverr, and TaskRabbit connect service providers with customers through digital platforms. These systems often rely on filters such as ratings, reviews, location, and service categories to help users identify suitable workers. Although these platforms provide convenient access to service providers, they typically depend on simple ranking algorithms rather than fully intelligent recommendation models that analyze multiple dynamic factors simultaneously.

Despite the progress made in existing research and commercial platforms, several limitations remain. Many systems rely heavily on historical interaction data, making them less effective for new users or newly registered workers. Additionally, many recommendation systems focus on single-dimensional factors such as ratings or past interactions, ignoring other important attributes such as worker availability, proximity, and real-time service demand. Furthermore, complex machine learning models may require significant computational resources and large datasets, which may not be feasible for smaller platforms.

Therefore, there is a need for a more balanced recommendation approach that combines multiple worker attributes such as skills, experience, availability, ratings, and location while maintaining computational efficiency. The proposed Intelligent Worker Recommendation

Engine aims to address these limitations by implementing a multi-factor recommendation model capable of providing accurate and practical worker recommendations within a smart service marketplace environment.

1.3 Research Gap

Although many studies have explored worker and task recommendation systems in crowdsourcing environments, several important limitations still exist in current approaches. Existing research mainly focuses on recommending tasks to workers in large online platforms rather than recommending the most suitable workers for a specific service request in real-world service marketplaces. As a result, many current methods cannot effectively support real-time worker matching for household maintenance services or similar on-demand service platforms.

One major limitation in existing systems is the heavy dependence on historical interaction data. Many recommendation algorithms such as collaborative filtering and matrix factorization rely on past worker-task interactions to generate recommendations. However, these approaches suffer from the data sparsity and cold-start problem, where the system struggles to recommend tasks or workers when there is insufficient historical data for new workers or new tasks. Studies show that recommendation models in crowdsourcing often face severe data sparsity challenges, which significantly affect the accuracy and reliability of recommendations.

Another limitation is that many current recommendation models consider only a limited set of factors when generating recommendations. For example, some systems primarily analyze worker preferences or past performance but fail to incorporate other important contextual factors such as worker availability, real-time location, or service urgency. Research indicates that many task recommendation systems do not effectively integrate human or contextual factors into their models, which reduces their ability to provide personalized and realistic recommendations in practical scenarios.

Additionally, several studies focus on sophisticated machine learning or graph-based models that require large datasets and complex computational resources. While these advanced models can improve recommendation accuracy, they may be difficult to implement in smaller platforms or real-time applications. For example, knowledge graph and deep-learning-based recommendation models require extensive feature extraction and training data, making them less suitable for lightweight service marketplace systems.

Another research gap relates to the dynamic nature of worker preferences and availability. Many existing recommendation systems assume that worker preferences remain constant over time. However, research shows that worker preferences and interests change frequently, and recommendation models that ignore these temporal changes may produce outdated or inaccurate recommendations.

Furthermore, current crowdsourcing systems primarily focus on task allocation in digital work environments rather than local service marketplaces. In real-world service platforms, additional

attributes such as geographic proximity, service category expertise, reputation scores, and job completion reliability must be considered simultaneously. Existing research rarely integrates all these attributes into a unified recommendation framework designed specifically for service provider selection.

To address these gaps, the proposed Intelligent Worker Recommendation Engine introduces a multi-factor recommendation framework specifically designed for service marketplace platforms. The system combines several attributes—including worker skills, experience, ratings, availability, and geographic proximity—to generate a comprehensive suitability score for each worker. Unlike many traditional approaches that rely solely on historical interaction data, the proposed model integrates both contextual and performance-based attributes to produce more practical and accurate recommendations.

The novelty of this research lies in its context-aware and multi-criteria worker matching approach, which is designed specifically for real-world household service environments rather than purely digital crowdsourcing tasks. By combining worker profile analysis, contextual information, and a weighted scoring recommendation model, the proposed system aims to improve recommendation accuracy while maintaining computational efficiency. This approach provides a practical solution for connecting service requesters with the most suitable workers, thereby addressing the limitations observed in many existing recommendation systems.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop an Intelligent Worker Recommendation Engine that analyzes multiple factors such as worker skills, experience, ratings, availability, and geographic proximity to recommend the most suitable workers for a user's service request within a smart service marketplace platform.

2.2 Specific Objectives

- **collect and structure worker and service request data**: including worker skills, job categories, experience levels, ratings, availability, and location information for use in the recommendation process.
- **design a multi-factor worker evaluation model**: that calculates a suitability score for each worker based on factors such as skill matching, distance, reliability, and past performance.
- **implement a recommendation algorithm**: that ranks workers according to their suitability scores and generates a list of the most appropriate workers for a given service request.
- **integrate the recommendation engine with the service marketplace system:** so that users can receive real-time worker recommendations when they submit a service request.
- **evaluate the effectiveness of the recommendation engine**: by measuring recommendation accuracy, response time, and user satisfaction in selecting appropriate workers for tasks.
- **improve the recommendation quality**: by analyzing user feedback and worker performance data to refine the recommendation model.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach. The reason for using DSR is that the project involves:

- Designing and building an artifact (the Intelligent Worker Recommendation Engine)
- Evaluating its performance in a practical environment
- Solving a real-world problem: matching service requesters with skilled workers efficiently

Unlike purely experimental research, DSR focuses on creating innovative solutions and evaluating their effectiveness in context.

3.2 Technical Methods and Algorithms

The Intelligent Worker Recommendation Engine uses a hybrid recommendation approach, combining:

1. **Random Forest Classifier (Machine Learning)**
    - Predicts worker suitability probability based on historical data
    - Input features: skill match, rating, experience, distance, availability, success rate
    - Output: probability of successful task completion
2. **Multi-Criteria Weighted Scoring**
    - Computes final recommendation score by combining:
        - ML predicted suitability
        - Worker ratings
        - Proximity
        - Experience
        - Availability
    - Provides explainable ranking for real-time recommendation

This hybrid approach ensures intelligent, practical, and transparent recommendations.

3.3 Tools and Platforms with Justification

Python - Widely used for Random Forest, supports pandas, scikit-learn, joblib

FastAPI - Lightweight, high-performance framework for serving recommendations

PostgreSQL / MySQL - Stores worker profiles, availability, historical data

React Native - Cross-platform development for Android and iOS

3.4 Data Collection and Ethical Considerations

Data Collection:

- Worker attributes: skills, ratings, experience, past jobs, availability, location
- User service requests: service type, location, urgency, preferred time

Sources:

- Simulated dataset for development and testing
- Historical data from service marketplaces (if accessible)

Ethical Considerations:

- Anonymization of worker data to protect privacy
- Consent if real data is used
- Secure storage and access control of data
- Transparency: users know how recommendations are generated

3.5 Validation Strategy and Performance Metrics

Validation Strategy:

- Split dataset into training (80%) and testing (20%)
- Train Random Forest on training set
- Test prediction on testing set
- Evaluate overall system using simulation of service requests

Performance Metrics:

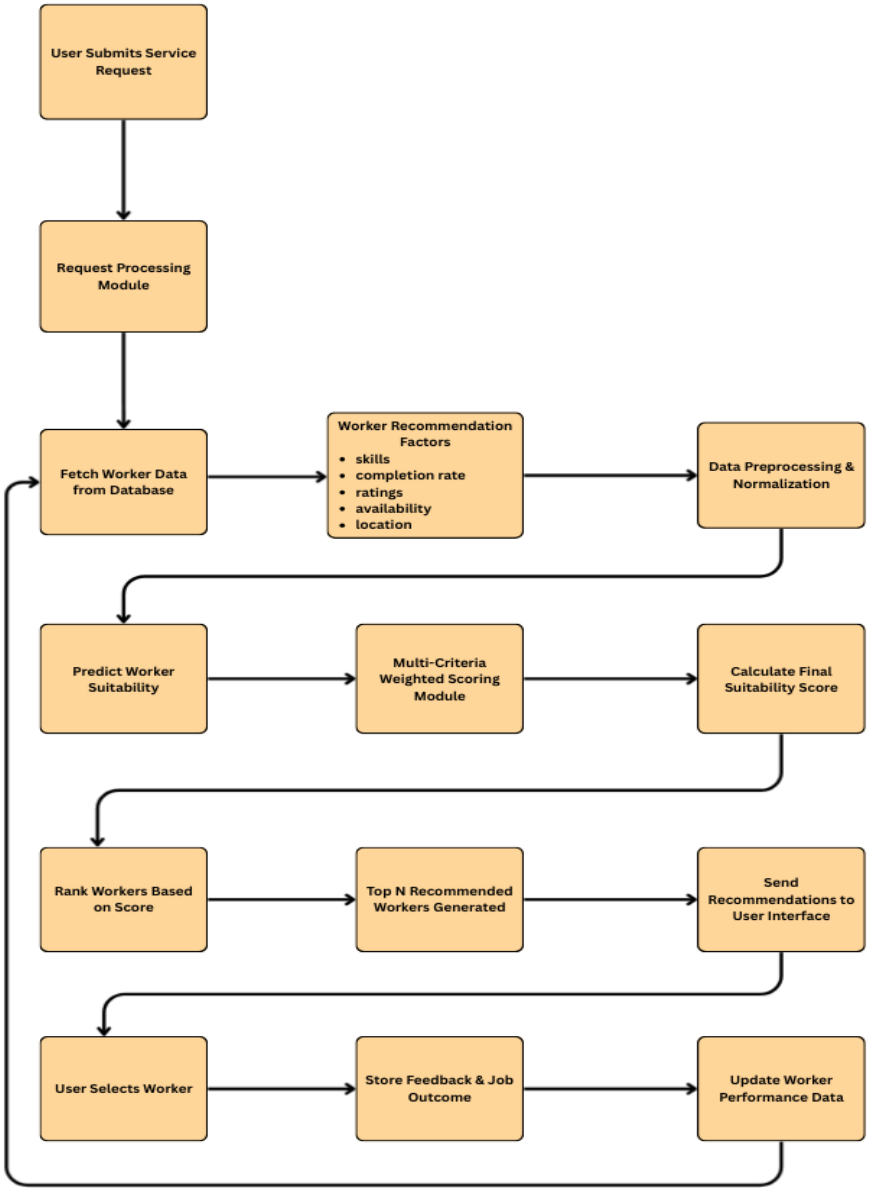
1. **Accuracy** – percentage of correct worker suitability predictions
2. **Precision & Recall** – evaluate correctness and completeness of recommendations
3. **F1 Score** – balance between precision and recall
4. **Top-N Accuracy** – percentage of times the actual suitable worker is in top-N recommendations

5. **Response Time** – how fast the system generates recommendations
6. **User Satisfaction** – feedback rating from users (optional in real implementation)

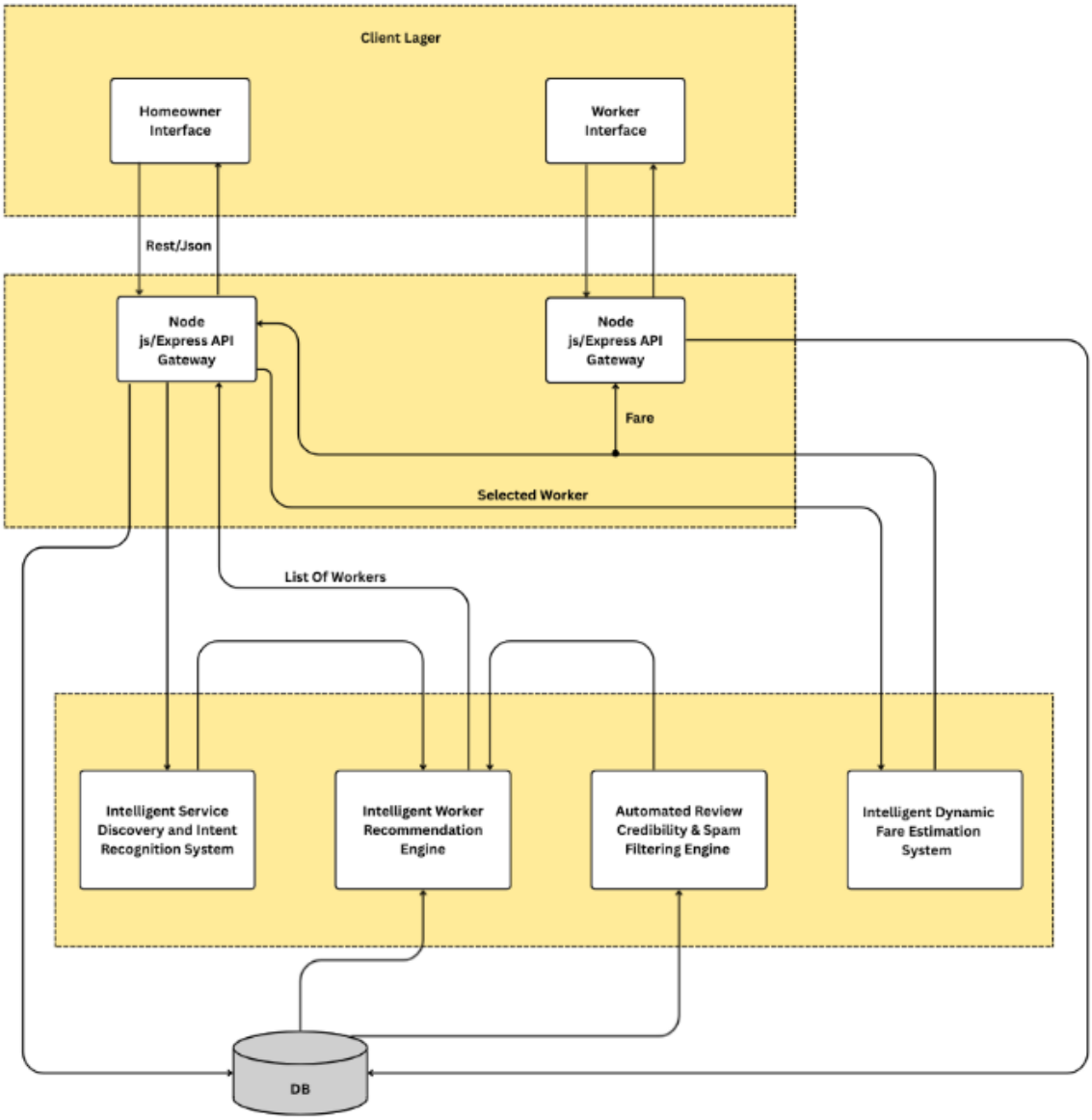
This combination ensures both ML model correctness and practical performance of the recommendation engine.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

- **Component specific system diagram**



- **System Architecture Diagram**



5. USER REQUIREMENTS

5.1 Stakeholder Identification

To accurately define the system requirements, the primary stakeholders interacting with the platform have been identified and categorized based on their roles and objectives:

1. **Homeowners (Service Seekers):** Non-technical users who require an intuitive mobile interface to report household maintenance issues using their natural, conversational language (English or Singlish) without navigating complex taxonomy menus.
2. **Skilled Workers (Service Providers):** Blue-collar professionals (e.g., plumbers, electricians, cleaners) who require a streamlined application to receive targeted job leads, optimized travel routes, and fair, dynamically calculated pricing for their services.
3. **System Administrators:** Backend operators responsible for managing the central Node.js platform, monitoring the performance of the AI microservices, handling user verification, and maintaining the MongoDB database.

5.2 Functional Requirements

1.Receive Service Request

The system shall receive and process service requests submitted by users, including details such as service type, location, urgency level, and preferred service time.

2.Extract Request Information

The system shall extract relevant information from the user request to identify required worker skills and contextual factors needed for recommendation.

3.Retrieve Worker Data

The system shall retrieve worker information from the database, including skills, ratings, experience, availability status, geographic location, and past job performance.

4.Perform Data Preprocessing

The system shall preprocess and normalize worker and request data to prepare feature inputs for the recommendation algorithm and machine learning model.

5.Generate Worker Suitability Prediction

The system shall use a machine learning model (e.g., Random Forest) to predict the suitability probability of each worker for the given service request.

6.Calculate Multi-Criteria Recommendation Score

The system shall calculate a final recommendation score for each worker using a weighted scoring mechanism that considers multiple factors such as skill match, distance, rating, experience, availability, and predicted suitability.

7.Rank Workers

The system shall rank workers based on their final recommendation scores in descending order.

8.Provide Top Worker Recommendations

The system shall generate and return a list of top recommended workers to the user interface for selection.

9.Capture User Selection and Feedback

The system shall record the worker selected by the user and collect feedback such as ratings, job completion status, and service satisfaction.

10.Update Worker Performance Data

The system shall update worker performance records based on job outcomes and feedback received from users.

11.Support Model Retraining

The system shall store historical interaction data to support periodic retraining and improvement of the recommendation model.

12.Provide Real-Time Recommendations

The system shall generate worker recommendations within an acceptable response time after receiving a service request.

5.3 Non-Functional Requirements

To ensure the platform is viable for real-world commercial deployment, the overall system architecture must adhere to the following non-functional quality attributes:

- **Performance and Latency:** Because user drop-off increases with wait times, the central Node.js API Gateway must return a complete match response to the mobile application in under 2.0 seconds. To support this, the independent Python AI microservices (such as Component 1) must execute their specific machine learning predictions in less than 200 milliseconds per request.
- **Scalability:** The system must utilize a decoupled microservices architecture. If the platform experiences a surge in user traffic, the system administrators must be able to horizontally scale the Node.js API Gateway and the individual Python ML containers independently across cloud infrastructure (e.g., AWS or Render) to prevent computational bottlenecking.
- **Security and Data Privacy:** All data transmitted between the mobile applications, the API Gateway, and the microservices must be encrypted in transit using HTTPS/TLS protocols. User authentication must be secured via stateless JSON Web Tokens (JWT). Furthermore, the overarching system must anonymize user problem descriptions before logging them in the MongoDB database to comply with standard data privacy regulations.
- **Usability and Accessibility:** The mobile application must be strictly "mobile-first" and designed for high accessibility. The system must possess a high tolerance for spelling errors and out-of-vocabulary phonetic variations, ensuring that users with lower technological literacy or informal typing habits can successfully operate the platform.
- **Reliability and Fault Tolerance:** The centralized system must aim for a 99.9% uptime availability. In the event that an individual AI microservice temporarily fails or times out, the central Node.js server must possess fallback mechanisms (e.g., gracefully degrading to a manual category selection menu on the user's app) rather than crashing the entire application.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The initial target market focuses on urban and suburban demographics within the Western Province of Sri Lanka, specifically targeting the informal household maintenance sector. The platform serves a dual-sided marketplace consisting of two primary customer personas:

- **Persona A (The Homeowner/Consumer):** Busy middle-to-high-income professionals aged 25–55. They value time and convenience but often lack the technical vocabulary to diagnose household faults. They predominantly communicate via mobile devices using conversational English or Singlish.
- **Persona B (The Service Provider/Worker):** Skilled independent contractors (e.g., plumbers, electricians, masons, and cleaners). They possess basic smartphone literacy but lack the capital for digital marketing. Their primary pain point is the inconsistency of daily work and the time wasted traveling to jobs that do not match their specific sub-skills.

6.2 Value Proposition

The platform's core value proposition is "Frictionless, Language-Agnostic Service Discovery."

- **For Homeowners:** It eliminates the cognitive load of navigating complex service directories. Users can simply type their problem as they would text a friend (e.g., "Mage AC eka wada na"), and the AI instantly bridges the communication gap, automatically dispatching the correct professional.
- **For Workers:** It provides a zero-marketing-cost pipeline of highly targeted jobs. Because the NLP engine maps symptoms directly to specific subcategories, workers only receive notifications for jobs they are explicitly qualified to perform, significantly increasing their conversion rates.

6.3 Revenue Model

The platform will operate on a **Commission-Based Transaction Model** combined with a freemium SaaS architecture to ensure rapid initial user acquisition:

- **Homeowner Usage:** 100% free to download and submit service requests.
- **Transaction Commission (Take Rate):** A standard platform fee (e.g., 10%–15%) is automatically deducted from the total dynamic fare calculated by Component 4 upon the successful completion of a job.
- **Worker Premium Tier:** While basic profile listing is free, workers can opt into a low-cost monthly subscription to unlock advanced analytics (e.g., viewing high-demand geographical zones) or priority ranking in Component 2's recommendation algorithm.

6.4 Cost Estimation

Transitioning the FYP prototype into a commercial Minimum Viable Product (MVP) will incur the following estimated costs (calculated for Year 1 of operations):

- **Capital Expenditure:**
    - Legal & Business Registration (incorporation, trademarking): ~$150
    - Initial Cloud Setup & AI Training Compute (Google Colab Pro / AWS EC2): ~$200
- **Operational Expenditure:**
    - Cloud Hosting Architecture (Node.js Gateway on Heroku/AWS, Python FastAPI microservices on Render): ~$80/month.
    - Database Hosting (MongoDB Atlas scaling tiers): ~$30/month.
    - Third-Party APIs (Google Maps API for Component 3 routing): ~$50/month (usage dependent).
    - Targeted Digital Marketing & Social Media User Acquisition: ~$150/month.

6.5 Pricing Strategy

The platform leverages a Dynamic Pricing Strategy (driven by Component 4). Instead of static catalog prices that ignore market realities, the system calculates fares based on real-time variables such as GPS distance, time of day, urgency of the request, and current worker availability. This ensures fair compensation for the worker during peak hours or emergencies while maintaining transparent, upfront cost estimations for the homeowner before they confirm the booking.

6.6 Competitive Advantage

Currently, the local market relies on fragmented Facebook groups, traditional word-of-mouth, or static classified websites. The platform's distinct competitive advantage is its Proprietary Intent Recognition Engine (Component 1). Existing competitors rely on rigid, exact-match English keyword searches, which fail when users input localized hardware slang or Singlish symptoms. By utilizing a bilingual SVM pipeline capable of processing unstructured code-mixed text, this platform captures a massive, unserved demographic that competitors mathematically cannot process.

6.7 Intellectual Property (IP) Considerations

To protect the commercial viability of the platform, several IP strategies will be implemented:

- **Data IP (The Moat):** The custom-built, annotated bilingual dataset mapping Singlish household problem descriptions to professional worker categories is highly unique. This proprietary dataset serves as a major barrier to entry for local competitors attempting to train similar AI models.

- **Codebase & Algorithms:** The specific text preprocessing pipeline (unified English-Singlish stop-word filtering) and the orchestration logic between the microservices will be protected under standard software copyright laws.
- **Brand Identity:** The platform's name, logo, and UI/UX design will be trademarked to establish brand trust in the local market.
- **Terms of Service:** Legal frameworks will be established within the app to strictly prohibit automated scraping of the platform's worker database or pricing algorithms by rival tech companies.

7. BUDGET AND JUSTIFICATION

The proposed budget is highly optimized for a university Final Year Project, ensuring that all funds are directed toward essential research and development infrastructure.

- **Internet Charges & AI Tools:** This allocates funds across all four group members over the 8-to-9-month project duration. Because the core of this project relies heavily on Artificial Intelligence, the team requires consistent, high-bandwidth internet access. This ensures members can seamlessly download large machine learning libraries (e.g., Hugging Face Transformers, Scikit-learn), access cloud-based training environments (like Google Colab), and utilize AI-assisted development tools for technical learning and debugging. Furthermore, stable connectivity is mandatory for the continuous integration and remote collaboration required by the microservices architecture.
- **Hosting:** This minimal hosting budget is designated for the final testing, integration, and presentation phases of the project (Phases 4 and 5). It covers the cost of deploying the Node.js API Gateway and the four independent Python FastAPI microservices to low-cost developer tiers on cloud platforms (such as Render or Heroku). This ensures the evaluation panel can interact with a live, functioning prototype of the mobile application and backend systems without incurring the high costs of commercial-grade server infrastructure.

Component	**Amount in USD**
Internet chargers (The development and technical information learning, AI tools)	30
Hosting	20
Total	50

8. WORK BREAKDOWN STRUCTURE (WBS)

7.1 Task Breakdown per Member

The project utilizes a microservices architecture, which naturally facilitates parallel development and prevents technical bottlenecks. The workload is strictly balanced by assigning each member ownership of one complex machine learning component, while standard full-stack development tasks are distributed equally across the team.

- Member 1: Bandara W.A.M.S. (IT22551320) – Intelligent Service Discovery and Intent Recognition System
    - **Core Responsibility:** Design and develop the NLP classification component.
    - **Specific Tasks:** Collect and preprocess a dataset of service queries; implement Intent Recognition algorithms to classify user problems; develop Entity Extraction logic to identify specific items like "Tap" or "Leak".

- Member 2: Shiwanthika M.M. (IT22283658) – Intelligent Worker Recommendation Engine
    - **Core Responsibility:** Design and develop the context-aware matching system.
    - **Specific Tasks:** Define weighted scoring metrics (Skill match, Distance, Education, Availability); develop the Hybrid Recommendation Algorithm; integrate real-time location data to calculate dynamic arrival time scores.

- Member 3: Wijebandara W.N.S.K. (IT22643704) – Automated Review Credibility & Spam Filtering Engine
    - **Core Responsibility:** Design and develop the trust and sentiment analysis component.
    - **Specific Tasks:** Collect a dataset of genuine and spam reviews for training; train a Sentiment Analysis model (e.g., Naive Bayes) to classify feedback; implement a verifiable "Trust Score" calculation based on review authenticity.

- Member 4: Kumara A.J.T.M. (IT22137982) – Intelligent Dynamic Fare Estimation System
    - **Core Responsibility:** Design and develop the predictive pricing component.
    - **Specific Tasks:** Conduct a market survey to establish baseline labor and transport rates; develop a Regression Model to predict repair costs based on job complexity; implement the 3-Stage Pricing Logic covering transport, inspection, and repair estimates.

- **Shared Team Responsibilities (Balanced Workload):** To ensure equitable distribution of labor, all members will collaborate on the following overarching platform requirements:

    - **Frontend Development:** Collaborative development of the Homeowner and Worker mobile/web interfaces utilizing React and Next.js.

- **Central Backend:** Co-developing the central Node.js/Express API Gateway and configuring the MongoDB database schemas.
- **Integration & Testing:** End-to-end API testing to ensure the Python microservices communicate seamlessly with the Node.js server.
- **Documentation:** Equal division of research proposal writing, interim reports, and the final thesis document.

7.2 Timeline Alignment

Phase	Milestone / Tasks	Duration	Primary Focus
Phase 1	**Project Planning & Architecture Design** Finalize proposal, define API contracts between microservices, set up cloud repositories, and design MongoDB schemas.	Months 1 - 2	Team Collaboration
Phase 2	**Data Collection & Core ML Development** Gather datasets, build preprocessing pipelines, train initial baseline machine learning models (NLP intent, Worker Ranking, Spam/Sentiment Analysis, Pricing Regression).	Months 3 - 4	Individual Components

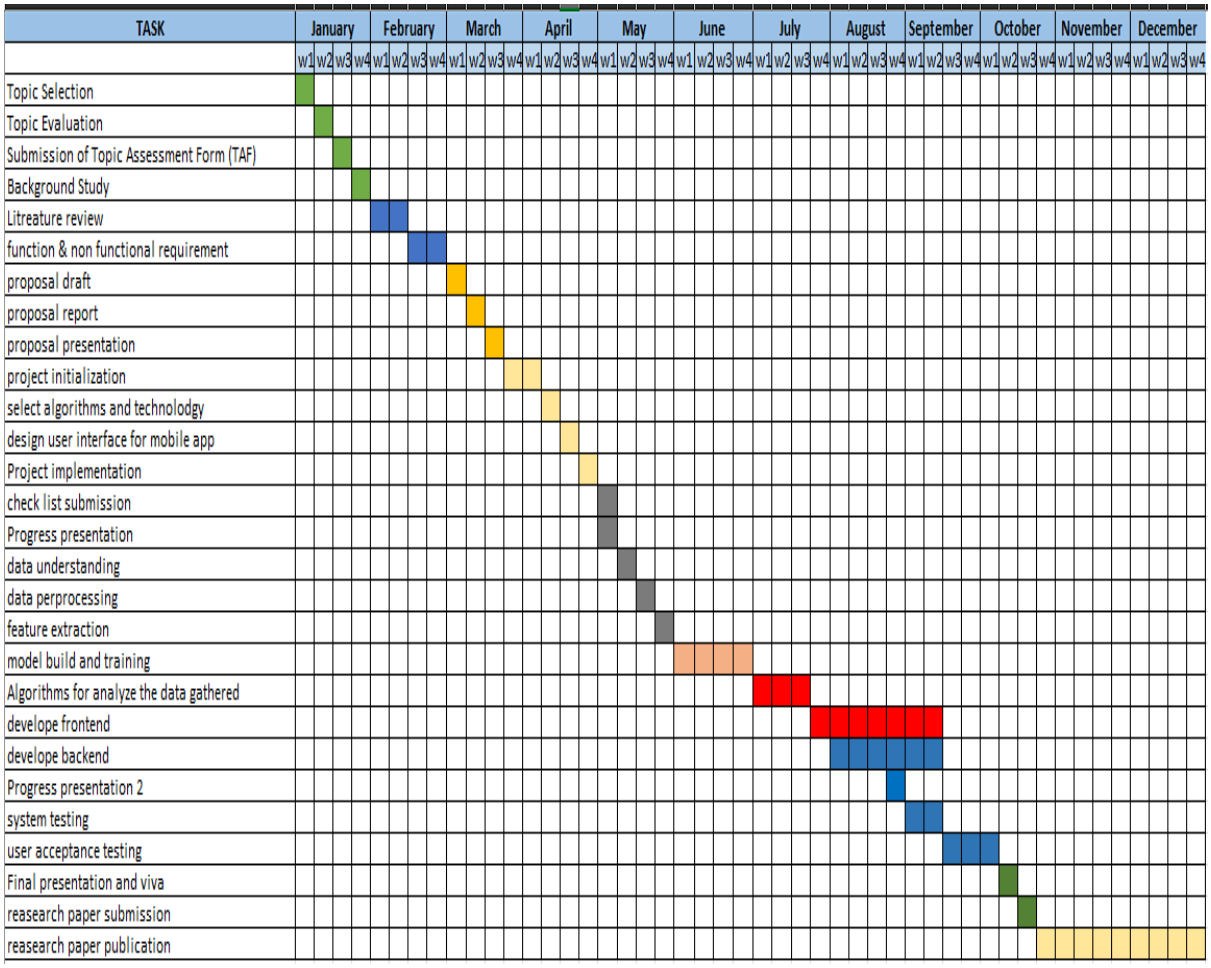
Phase 3	**Backend APIs & Frontend UI/UX** Wrap trained ML models in FastAPI, develop the Node.js API Gateway, and build the React/Next.js frontend user interfaces.	Months 5 - 6	Parallel Development
Phase 4	**System Integration & Rigorous Testing** Connect the frontend to the backend, ensure data flows correctly to all four AI microservices, conduct latency and accuracy testing.	Month 7	Team Collaboration
Phase 5	**Deployment, Evaluation & Documentation** Deploy the MVP to cloud infrastructure, finalize evaluation metrics, and complete the final academic thesis.	Months 8 - 9	Finalization

7.3 Workload Distribution Strategy

The workload distribution strategy is inherently balanced due to the decoupled nature of the system architecture. During Phase 2, all four members will face similar cognitive loads as they research, train, and test their respective machine learning models in isolation (e.g., Member 1 training SVMs, Member 3 training Naive Bayes). Because the components are separated by strict API endpoints, no member is blocked by another's progress. During Phases 3 and 4, the workload shifts from individual research to collaborative software engineering, ensuring all members contribute equally to the final deployable product.

9. GANTT CHART

- **Insert Gantt chart here (Semester-wise breakdown with milestones).**



10.REFERENCES LIST

[1] Y. Tong, L. Chen, and C. Shahabi, *"Task Recommendation in Crowdsourcing Based on Learning Preferences and Reliabilities,"* arXiv preprint arXiv:1807.10444, 2018.

[2] J. Yang, Y. Tong, C. Shahabi, and L. Chen, *"Context-Aware Hierarchical Online Learning for Performance Maximization in Mobile Crowdsourcing,"* arXiv preprint arXiv:1705.03822, 2017.

[3] A. Nikzad-Khasmakhi, M. Balafar, and M. Feizi-Derakhshi, *"From Zero to the Hero: A Collaborative Market-Aware Recommendation System for Crowd Workers,"* arXiv preprint arXiv:2107.02890, 2021.

[4] K. Ong, S. C. Haw, and K. W. Ng, "Deep Learning Based Recommendation System: An Overview on Models, Datasets, Evaluation Metrics, and Future Trends," *ACM International Conference Proceeding Series*, pp. 6–11, 2019. Available: https://eurekamag.com/research/102/791/102791362.php

[5] X. Wei, H. Zhang, and Y. Li, *"A Crowdsourcing Recommendation that Considers the Influence of Workers,"* Computers, Materials & Continua, vol. 66, no. 2, pp. 1887–1902, 2020.

[6] M. Xu, Z. Chen, and Y. Wang, *"GRBMC: An Effective Crowdsourcing Recommendation for Worker Groups,"* Expert Systems with Applications, vol. 186, 2021, Art. no. 115039.

[7] A. Nikzad-Khasmakhi, M. Balafar, and M. Feizi-Derakhshi, *"Role-Based Clustering for Collaborative Recommendations in Crowdsourcing Systems,"* in Proc. Int. Conf. on Intelligent and Fuzzy Systems, 2019, pp. 245–253.\

[8] S. Sharma et al., "Machine Learning-Based Recommendation Systems: A Survey," *Knowledge-Based Systems*, vol. 215, 2021. Available: https://www.sciencedirect.com/science/article/pii/S0950705121000332

[9] [3] G. B. N. Jyothi, S. C. Gurram, and K. Mounika, “Product Recommendation System Using Machine Learning Techniques,” *Lecture Notes in Networks and Systems*, 2026. Available: https://eurekamag.com/research/102/934/102934019.php

[10] S. R. Babu and R. B. Shettar, *“Task Recommendation in Reward-Based Crowdsourcing Systems,”* in Proc. IEEE Int. Conf. on Computational Intelligence and Computing Research (ICCIC), Coimbatore, India, 2017, pp. 1–5.

11.APPENDICES



